
A typical flatbed scanner

Another category is the rotary scanner, used for high-speed document scanning. This is another kind of drum scanner, but it uses a CCD array instead of a photomultiplier. Other types of scanners are planetary scanners, which take photographs of books and documents, and digital camera scanners, which are based on the concept of reprographic cameras. Due to increasing resolution and new features such as anti-shake, digital cameras have become an attractive alternative to regular scanners. While still containing disadvantages compared to traditional scanners, digital cameras offer unmatched advantages in speed and portability.

There are different types of scanners, depending on the user's purposes. Described below are the most commonly used types that can be found in the market:

Drum scanners capture image information with photomultiplier tubes (PMT), rather than the charge-coupled-device (CCD) arrays found in flatbed scanners and inexpensive film scanners. Reflective and transmissive originals are mounted on an acrylic cylinder, the scanner drum, which rotates at high speed while it passes the object being scanned in front of precision optics that deliver image information to the PMTs. Most modern colour drum scanners use 3 matched PMTs, which read red, blue, and green light respectively. Light from the original artwork is split into separate red, blue, and green beams in the optical bench of the scanner.

The drum scanner gets its name from the large glass drum on which the original artwork is mounted for scanning—they usually take 11x17-inch documents, but the maximum size varies with the manufacturer. One of the unique features of drum scanners is the